AI Assisted Coding

Lab Assignment-19.1

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Batch:14

TASK 1 – Python to Java Conversion

Prompt:

Convert the following Python class into an equivalent Java class. Ensure constructors, attributes, and methods are properly translated.

Code:

```python
class Student:
    def __init__(self, name, marks):
        self.name = name
        self.marks = marks
    def display(self):
        print("Name:", self.name)
        print("Marks:", self.marks)
s = Student("Ravi", 85)
s.display()
```

Code:

```java
public class Student {
    private String name;
    private int marks;
    public Student(String name, int marks) {
        this.name = name;
        this.marks = marks;
    }
    public void display() {
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }
    public static void main(String[] args) {
        Student s = new Student("Ravi", 85);
        s.display();
    }
}
```

Output:

```
Name: Ravi
Marks: 85
```

Explanation:

- Python uses dynamic typing, while Java requires explicit data types.
- The Python constructor __init__becomes a Java constructor with the class name.
- Java requires a main() method to run the program.

TASK 2 – C++ to Python Function Conversion

Prompt:

Translate the following recursive C++ factorial function into Python using Pythonic conventions.

C++ Code:

```cpp
#include <iostream>
using namespace std;
int factorial(int n) {
    if (n == 0 || n == 1)
        return 1;
    return n * factorial(n - 1);
}
int main() {
    cout << factorial(5);
    return 0;
}
```

Python Code:

```python
def factorial(n):
    if n == 0 or n == 1:
        return 1
    return n * factorial(n - 1)
print(factorial(5))
```

Output:

```
120
```

Explanation:

- C++ uses type declarations (int), Python does not.
- ‘||’ in C++ becomes ‘or’ in Python.
- Python does not need a main function or semicolons.

TASK 3 – SQL to Pandas Query

Prompt:

Convert the following SQL query into an equivalent Pandas DataFrame operation.

SQL Code:

```sql
SELECT name, salary
FROM employees
WHERE salary > 50000;
```

Pandas Code:

```python
import pandas as pd
data = {
    "name": ["Asha", "Ravi", "Kiran", "Meena"],
    "salary": [45000, 60000, 52000, 48000]
}
df = pd.DataFrame(data)
result = df[df["salary"] > 50000][["name", "salary"]]
print(result)
```

Output:

```
   name  salary
1   Ravi   60000
2  Kiran   52000
```

Explanation:

- SQL queries operate on database tables, while Pandas operates on DataFrames.
- The WHERE clause becomes boolean filtering in Pandas.
- Column selection is done using double brackets [["col1", "col2"]].

TASK 4 – Real-Time Application: Multi-Language Snippet Converter

Prompt:

Convert the following Python web scraping code into equivalent JavaScript for browser automation.

Python Code:

```python
import requests
from bs4 import BeautifulSoup
url = "https://sraap.in/"
response = requests.get(url)
soup = BeautifulSoup(response.text, "html.parser")
for link in soup.find_all("a"):
    print(link.get("href"))
```

JavaScript Code:

```javascript
const links = document.querySelectorAll("a");
links.forEach(link => {
    console.log(link.href);
});
```

Output:

```
index.php
staff_login.php
student_login.php
```

Explanation:

- Python uses external libraries (Requests, BeautifulSoup) for scraping.
- JavaScript in the browser directly accesses the DOM.
- Syntax differs: Python uses indentation, JavaScript uses braces {}.
- Execution environments are different (Python = server, JS = browser).